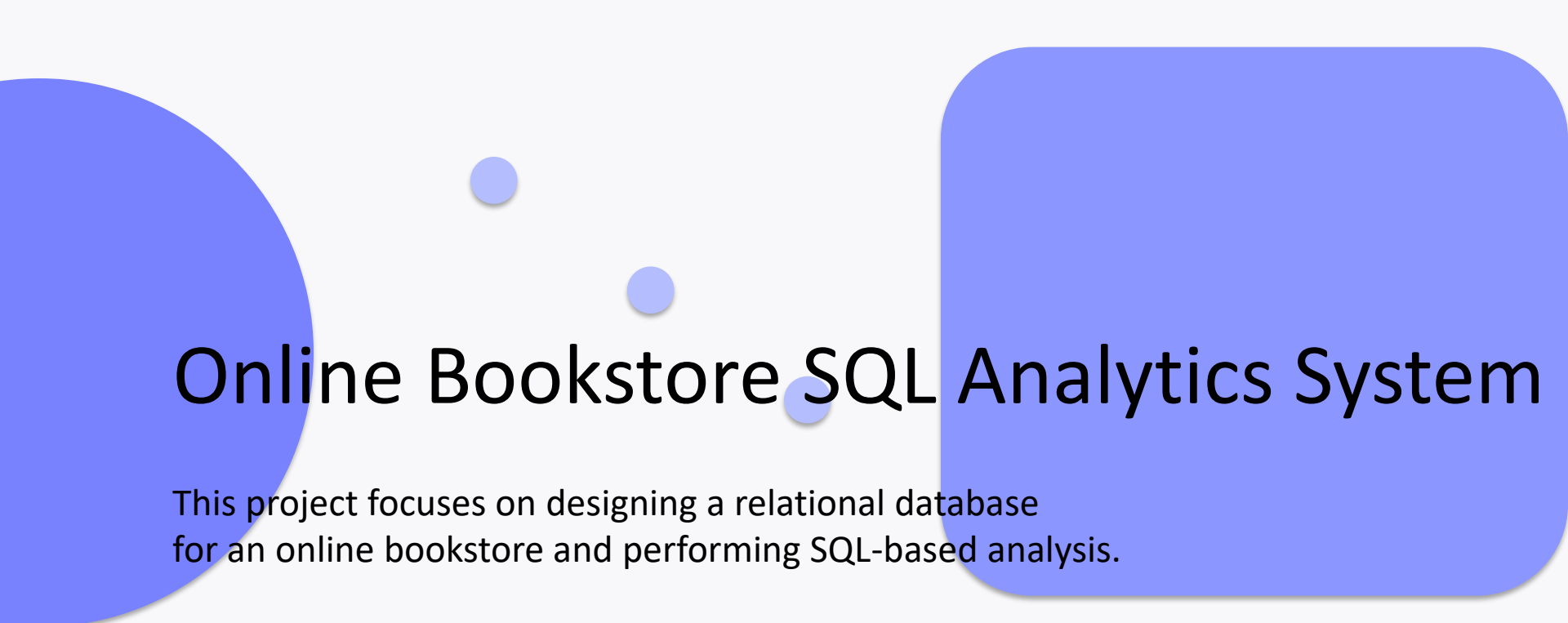
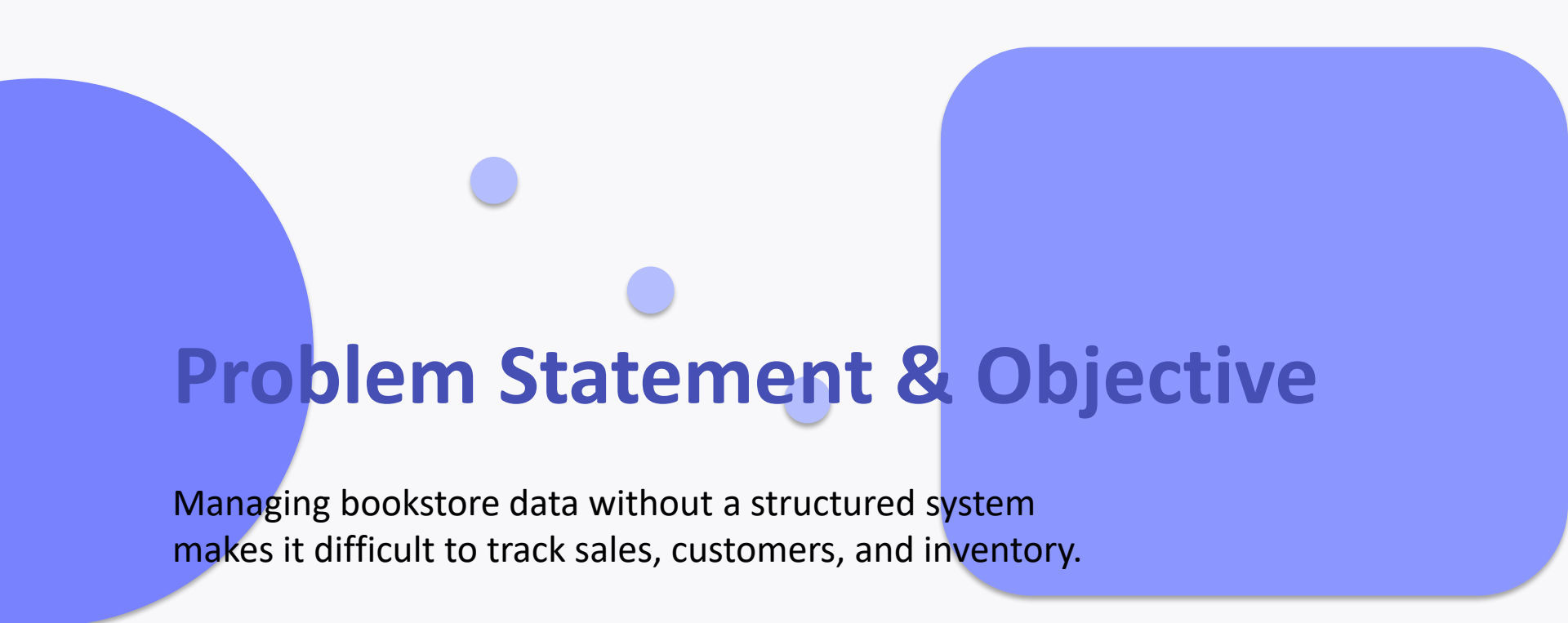




Online Bookstore SQL Analytics System

This project focuses on designing a relational database for an online bookstore and performing SQL-based analysis.

- Structured 3 interconnected tables (Books, Customers, Orders)
- Processed 1000+ records
- Executed 15+ SQL queries
- Generated insights on sales, customers, and inventory



Problem Statement & Objective

Managing bookstore data without a structured system makes it difficult to track sales, customers, and inventory.

Objective:

- Design a relational database
- Establish table relationships
- Perform analytical queries
- Extract meaningful business insights

Retrieve all books in the "Fiction" genre

```
SELECT * FROM Books
WHERE genre = 'Fiction';
```

book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
31	Implemented encompassing conglomerat...	Melissa Taylor	Fiction	2010	21.23	44

Show orders placed in November 2023

```
SELECT * FROM Orders
WHERE Order_Date BETWEEN '2023-11-01' AND '2023-11-30';
```

order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
4	433	343	2023-11-25	7	301.21
19	496	60	2023-11-17	9	316.26
75	291	375	2023-11-30	5	170.75
132	469	333	2023-11-22	7	194.32
137	474	471	2023-11-25	8	363.04

Retrieve the total stock of books available

```
SELECT SUM(stock) AS Total_Stock  
FROM Books;
```

total_stock	
bigint	
	25056

Retrieve the total number of books sold for each genre

```
SELECT b.genre, SUM(o.quantity) AS Total_Books_Sold
FROM Orders o
JOIN Books b
ON o.book_id = b.book_id
GROUP BY b.genre;
```

genre character varying (50) 🔒	total_books_sold bigint 🔒
Romance	439
Biography	285
Mystery	504
Fantasy	446
Fiction	225
Non-Fiction	351
Science Fiction	447

List customers who have placed at least 2 orders

```
SELECT o.customer_ID, c.name, COUNT(o.order_ID) AS Order_Count
FROM Orders o
JOIN Customers c
ON o.customer_ID = c.customer_ID
GROUP BY o.Customer_ID, c.name
HAVING COUNT(Order_ID) >= 2;
```

customer_id integer	name character varying (100)	order_count bigint
225	Christopher Mccullough	2
418	Kiara Blankenship MD	3
322	William Cameron	3
325	Emily Vargas	4
376	Justin Donaldson	2

Find the most frequently ordered book

```
SELECT o.book_id, b.title, COUNT(o.order_id) AS Order_Count
FROM Orders o
JOIN Books b
ON o.book_id = b.book_id
GROUP BY o.book_id, b.title
ORDER BY Order_Count DESC
LIMIT 1;
```

book_id integer	title character varying (100)	order_count bigint
88	Robust tangible hardwa...	4

Retrieve the total quantity of books sold by each author

```
SELECT b.author, SUM(o.quantity) AS Total_Books_Sold
FROM Orders o
JOIN Books b
ON o.book_id = b.book_id
GROUP BY b.author;
```

author character varying (100) 	total_books_sold bigint 
Jared Cortez	10
Tracy Parker	11
Taylor Wang	9
Cathy Knight	6
Bianca Matthews	3

List the cities where customers who spent over \$30 are located

```
SELECT DISTINCT c.city, o.total_amount
FROM Orders o
JOIN Customers c
ON o.customer_id = c.customer_id
WHERE o.total_amount > 30;
```

city character varying (50) 🔒	total_amount numeric (10,2) 🔒
Taylorfurt	189.45
Leeport	141.39
Port Jasonview	149.12
Port Aaronstad	145.44
Matthewfurt	328.50

Find the customer who spent the most on orders

```
SELECT c.customer_id, c.name, SUM(o.total_amount) AS Total_Spent
FROM Orders o
JOIN Customers c
ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.name
ORDER BY Total_Spent DESC
LIMIT 1;
```

customer_id [PK] integer	name character varying (100)	total_spent numeric
457	Kim Turner	1398.90

Calculate the stock remaining after fulfilling all order

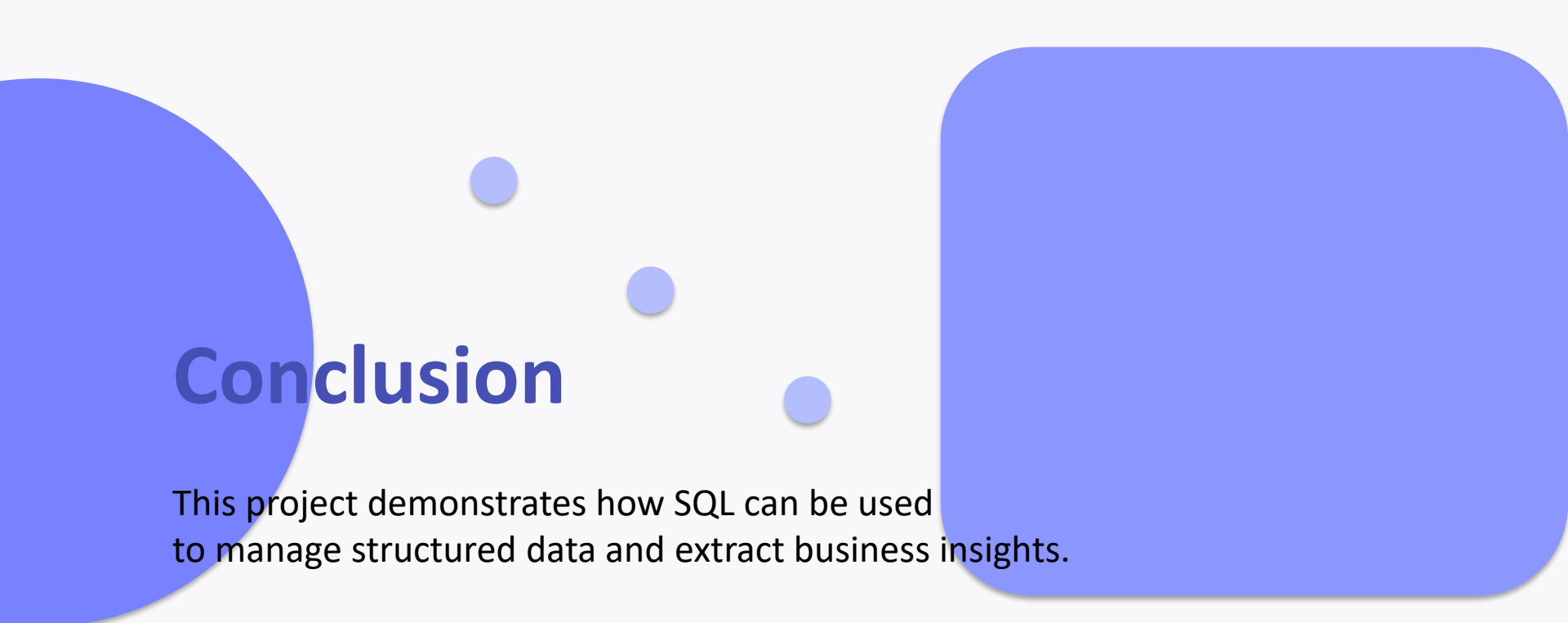
```
SELECT b.book_id, b.title, b.stock, COALESCE(SUM(quantity),0) AS Order_Quantity,  
       b.stock - COALESCE(SUM(quantity),0) AS Remaining_Quantity  
FROM Books b  
LEFT JOIN Orders o  
ON b.book_id = o.book_id  
GROUP BY b.book_id  
ORDER BY b.book_id;
```

book_id [PK] integer	title character varying (100)	stock integer	order_quantity bigint	remaining_quantity bigint
1	Configurable modular throughput	100	3	97
2	Persevering reciprocal knowledge user	19	0	19
3	Streamlined coherent initiative	27	5	22
4	Customizable 24hour product	8	0	8
5	Adaptive 5thgeneration encoding	16	8	8



Key Insights

- Top-selling books contribute major revenue
- Few customers drive most sales
- Inventory tracking improves efficiency
- SQL enables data-driven decisions



Conclusion

This project demonstrates how SQL can be used to manage structured data and extract business insights.

It highlights:

- Database design
- Query optimization
- Data-driven decision making